Putting Things in Place: Institutional Objects and Institutional Logics

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Abstract

This paper explores the role of institutional objects in the constitution of institutional logics. Institutional objects depend for their objectivity on the goods produced through those objects, such as economic models, passports or sacred texts. We theorize institutional logics as grammars of valuation that institutionalize goods through institutional objects. We identify four value moments through which goods are objectified: *institution*, the instituting of a good, a belief and an imagination of its objective goodness; *production*, how the good is produced, what practices are productive of the good; *evaluation*, how good is the good, the practices and objects through which worth in terms of that good is determined, and *territorialization*, the domain of reference of the good, to what objects and practices a good can and does refer in its instantiations. We assess the adequacy of our model through an institutional object based on the good of "market value" - i.e. an options pricing model. We discuss the implications of these findings for institutional logical theory and the sociology of valuation.

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“We live in capitalism. Its power seems inescapable. So, did the divine right of kings."
Ursula Le Guin, 2014 National Book Award

Introduction

The institutional logical approach has tended to be immaterial and valueless. In this essay, we seek to take conceptual steps to make institutional logics both material and moral, and indeed necessarily both at the same time. Institutional logics are “supraorganizational patterns of activity by which humans conduct their material life in time and space, and symbolic systems through which they categorize that activity and infuse it with meaning.” (Friedland and Alford 1991, p. 243). An institutional logic is not a logical relation of propositions grounded in presumption of representation; it is a productive logic of practice. An institutional logic is a socially meaningful form of material practice (Friedland 2017a, Smets et al. 2012). Its materiality is given by the objects deployed in practice and the corporeality of the practitioners. Its meaning is given by the non-phenomenal good presumed, produced, evaluated and instantiated by those practices. Such goods include justice, security, sovereignty, democracy, individuality, knowledge, information, health, romantic love, God, art, nature, or market value.[1]

Institutional logics imply a relational approach to the pathways of practice that join persons and objects to each other, akin to accounts of social life made by relational anthropologists and sociologists (Mol 1999, 2002, Zelizer 2012). The question is whether the primary elements of an institutional logic cohere. The basis of that coherence cannot be adequately explained by the attributes of the elements which can be additively netted out in a general linear model, but as relational wholes, not unlike the way Martin describes the elements of fields as “nodes in sets of relations,” “defined by their reciprocal positions” (Martin 2011, p. 241). Although conceptualized as a genre of socio-materiality, in which social practices and objects are co-constitutive (Jones et al. 2020, Orlikowski 2007, Smets et al. 2015), the objects in and through which that practice is carried out have been largely absent (Boxenbaum et al. 2016, 2018, Jones et al. 2017). The approach has been weighted towards actions or activities, to practices given sense and classified

[1] The goods enumerated here are not systematically derived from a theory of society, but are some of the dominant goods evoked in practice in modern societies that come to mind. An institutional logical approach, at this point, is being built on a piecemeal basis.

by subjects through available codes and schema, “vocabularies of practice” Thornton, Ocasio and Lounsbury (Thornton et al. 2012, p. 94) call them, tending to marginalize the objects through which these are accomplished.

And although an institutional logic was conceptualized as an inherently evaluative modality, the approach has not taken up the ways in which values are related to the objects through and towards which practices are organized (Friedland 2012). Value was central to ‘old’ institutionalism; it has been marginalized in the ‘new’ with its categorical, cognitive and causal emphases (Friedland 2017b, Greenwood et al. 2002, Krygier 2015). The institutional logical literature has been oriented towards classification and categorization of objects, towards what something is taken as in a kind of social constructionist nominalism. This is, of course, warranted by the centrality of modalities of language-use in the co-constitution of the elements of an institutional logic.[2] But it has tended to efface the teleological, the good or values variously named, believed or immanent in practice (Friedland 2012). Institutional logical work, in stark contrast to the conventions of worth school, has neglected the good as a constituent, indeed a cause, of that practice, and how objects and subjects conjointly sustain and are sustained by that good. Without value, of what a good is and what is good, institutional logics are left without purpose, with a reductive account of what animates social life.

Institutional logics do not seek an account of individual practices, but the grammars of practice available to practitioners. Grammar traditionally referred to the rules of language, in the sense of the syntactical rules of language independent of semantic meaning. Our use of the concept of grammar is akin to that of Anglo-Austrian philosopher Ludwig Wittgenstein for whom grammar describes the systematic way that the meanings of words are conditioned by their normative, context-dependent uses. Beyond the use of words themselves, words are embedded in practices and vice versa. Given that practices take place through objects, grammar is also essential to the ontological constitution of objects. “Grammar tells what kind of object anything is.” (Wittgenstein 1953, p. 373). The grammar is embedded in given forms of practice, or as he calls them, “forms of life” (Wittgenstein 1953, p. 241).[3] “The speaking of language is part of an activity, or of a form

[2] Schatzki remarks regarding Wittgenstein’s concept of a “pattern of life”: “The patterns that rely heavily on social practices, in the sense that people come to perform their constituent behaviors through social training and learning, almost always have correlative concepts” (Schatzki 1996, p. 32).

[3] Sparely used in *Philosophical Investigations,* the term “forms of life” has occasioned considerable debate as to whether it refers to a human form of life, or to cultural forms of human life. For Wittgenstein, it arguably applied to

of life." (Wittgenstein 1953, p. 23). The meaning of language is a constituent of that form of life. "To imagine a language means to imagine a form of life" (Cavell 1996, Moyal-Sharrock 2015, p. 33, Wittgenstein 1953, p. 19). It is agreement "in the language they use" that grounds people's saying what is true (Wittgenstein 1953, p. 88). The saying of true and false is "not agreement in opinions, but in forms of life." Words and objects are here joined not through representation, but through their co-constitutive use in modalities of practice.

Luc Boltanski and Laurent Thévenot, dubbed French pragmatists, also use the category of grammar in their celebrated text *On Justification* to describe the "grammar of political bonds," based on "different principles of order," on the one hand, common goods that give us "in what the worth of the worthy consists" and a coherent "set of objects" associated with those principles that afford "natural situations" in which "tests" of worth take place that produce "a justifiable order among persons" (Boltanski and Thévenot 2006, p. 13–14, 40, 228).[4] Grammars here refer to the constraints to which "legitimate forms of agreement" must conform to justify contested evaluations of, and by implication distributions, of worth (Boltanski and Thévenot 2006, p. 39).

Beyond the scope differences between the two grammars, that the first is general and granular, and the second political and typological, thinking the difference between them helps identify the theoretical space in which an institutional logic as good-dependent practice is situated. Wittgenstein's grammars examine the inner workings of practices, including of course the practice of language use, which cohere into "forms of life." He does not point either to a representational human mind, nor a metaphysical ground behind or underneath the phenomenal operation of an order of life to which a language game is integral. Boltanski and Thévenot's orders of worth work with metaphysical common goods about which people agree which provide the principles by which one can achieve "the establishment of an acceptable proof," and objects either delimit or signal "zones of pertinence" where the good is a legitimate common good, or serve as evidence of the worthiness of a person in terms of that good, objects "to measure themselves against one another." (Boltanski and Thévenot 2006, p. 12, 40, 141, 151).

The core practices of an institutional logic are grounded in and ground particular forms of common good (see Figure 1). The existence, the meaning and the effects of these practices depend

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both (Moyal-Sharrock 2015, Schatzki 1996, p. 67). It is a term that stands at the horizon of his work, not something he specified.

4 The use of « grammar » here refers to Adam Smith's argument that « the rules of justice can be compared to the rules of grammar » (Smith 1759, p. 175), cited in Boltanski and Thévenot (2006, p. 44).

on the good. And reciprocally the good is constituted and known through these practices. Institutional logics with their emphasis on modalities of practice, rather than practitioners who evaluate or are evaluated, offer a way to restore value not only as an individual belief in, an identification with and a desire for a good located in the subject, as a subjectification, but as a purpose embodied in the objects and the practices they afford, an objectification. As grammars of good-dependent practices, institutional logics fuse fact and value. In an institutional logic, the relationship between the materiality of practice and the good is internal, not external; a material symbolization, not a sign; a performative, not a representation.[5]

Some objects are integral to an institutional logic; their core practices do not obtain without them, as in paper ballots and ballot boxes on the one hand and secret voting on the other. In this essay, we want to take up these kinds of objects whose nature and effects in practice depend on a belief in a particular good -- representation or popular will in this example -- which is both integral to their objectivity and which their use is presumed to produce. We call these institutional objects. Such objects include money, contracts, property, territorial borders, passports, constitutions, communion wafers, audits, accounts, and financial models, among others. Most objects are not institutional; they are "already there" (Martin and Lembo 2016, p. 25); they do not depend for their existence and their effects on non-phenomenal goods. They can be understood in materialist terms, in the way things work, as intentional techniques.

Institutional objects are different. That such objects appear at all is contingent upon the content of their significance, on what a good is, on what is good and how good it is. A good here is both ontological, a social fact, an "isness," and teleological, a value or purpose, an "oughtness."[6] Institutional logics are good dependent practices: on the one side, an instrumental regime, a practical organization of meaningful means to an end, an end immanent in the means themselves, constitutive of the objects and the practices they afford, and on the other, a belief and investment

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[5] The internal, co-constitutive relationship between telos, practice and institutional objects is one among the many differences with the tradition of alignment in organizational theory, with which this approach has a surface kinship. Sender writes: "Organizational alignment is a descriptive concept referring to the extent to which the strategy, structure, and culture of the organization combine to create a synergistic whole that makes it possible to achieve the goals laid out in the organization's strategy" (Sender 1997, p. 27). In this account, for example, values are understood as subjective preferences as to what is good and bad, held in common as a shared culture (Sender 1997, p. 30). Values externally condition goals, tactics and norms, the last we would identify with the regularities of practice. The alignment of values with strategy and structure is what makes an organization efficient.

[6] It is to overcome this duality of fact and value, object and subject, that Friedland conceptualized this good as an institutional substance, using the Aristotelian category from his *Metaphysics*, what makes matter a "this," which points to an immanence in the object world, or an inherence as Spinoza would later term it (Friedland 2017b, a).

in a value – variously tacit or discursive, whether named or going without saying, that is the objective of the practices towards which these means are deployed. It is the conjunction of the two – of the object and the objective, of the ontological and the teleological, – that constitutes an institutional logic, both the object and the practices it affords.

While institutional logical theory has largely ignored the role of objects, the sociology of valuation has not (Fourcade 2011b, Hutter et al. 2015, MacKenzie and Millo 2003). Scholars in this tradition have investigated the mechanisms through which objects such as market devices like financial models produce the value they presume (Callon et al. 2007, Callon and Muniesa 2005). They have variously shown that the "process of valuation" – the process through which market devices shape what is deemed of value and how to assess this value – can be performative, not just representative (Helgesson and Muniesa 2013, MacKenzie 2006). Although the sociology of valuation has pointed to the constitutive role of institutional context (Fourcade 2011a, MacKenzie 2011), research bridging institutional logical theory and valuation has been scarce (Beunza and Ferraro 2019). The lack of specification of the institutional conditions that enable the performativity of market devices, for example, has led several to question the generalizability of such theory (Beunza and Ferraro 2019, Marti and Gond 2018).

We understand an institutional logic as a grammar of valuation, values objectified in practice[7]. Here we seek to specify the practices through which institutional objects such as market devices, institute, produce, evaluate and instantiate the goods that are the purpose of the practices they presume. An institutional logic is a work continuously in process. We identify four value moments: *institution*, the instituting of a good, a belief in and an imagination of its objective goodness as actualizable in practice; *production*, how the good is produced, what practices are productive of the good; *evaluation*, how good is the good, the practices and objects through which worth in terms of that good is determined, and *territorialization*, the domain of reference of the good, to what objects and practices a good can and does refer in its instantiations. The value moments phenomenonalize the good in practice and are co-constituting. These four value moments together articulate the grammar of practice, rendering the institutional logic coherent, legible and

[7] See Gehman, Treviño and Garud (2013) for an attempt at restoring values in practice through a performativity lens. While the authors focus on the processes through which organizational values come to be practiced in organizations, this essay focuses on the process of valuation attached to an institutional logic. Both papers share a desire for restoring values in the account of social life.

thereby institutionalized.[8] Taken for grantedness, typically posited as an institution's essence, is an effect of this grammar. An institutional logic is therefore a grammar of *valuation practices*, the regularities and rules, or conventions, by which these moments of valuation are constituted and cohere, their internal and external relations, the conditions and consequences of their efficacy. Although we are far from being able to specify, identify, operationalize and estimate the workings of that grammar, let alone the multiplicity of the goods involved and the way these grammars do or do not articulate, we offer these four value moments as necessary elements of any grammar of institutional practice. They are the moments through which a good is objectified, a value becomes a fact, a good becomes a real.

We examine the utility of this model by applying it to institutional objects grounded in and grounding the good of market value – known in practice through the monetary amount for which something can be sold in a market. Such institutional objects include relatively new market devices like financial models and accounting metrics, as well as older ones like fiat money. To do so, we re-read work in the sociology of valuation literature that explored the performativity of new market devices, recognized as having an increasing role in our everyday life (Davis 2010). To begin, we outline what institutional objects are. Then, we review the literature on the role of objects in the institutional logical literature and of institutional logics in the sociology of valuation. Next, we offer a theoretical model of the four value moments and the role of institutional objects therein. We then elaborate on MacKenzie and Millo's (2003) analysis of the performativity of an options-pricing model which we use to assess our model. We conclude by considering the explanatory and methodological implications of our model for institutional logical theory and the sociology of valuation.

Insert Figure 1 about here

[8] We are indebted to Sarah Friedland for this conceptualization of the grammar as an articulation of its practices. Articulation here is used in the sense of specifying distinct elements and their relationships to each other.

The Sociality and Materiality of Objects

What is an Object?

There is no unified conceptualization, let alone a theory, of the "object." As Pinch remarks, the word is "elusive and immediately problematic" (Pinch 2008, p. 467). Before considering the role of institutional objects, we need to clarify what we think an object is.[9]

First, an object is literally what is 'put before' the subject; it comes from *ob* – against – and *jacere* – to throw. An object is known and constructed vis-à-vis the human body – as a thing outside.[10] It is this exteriority and material resistance that makes an object independent from the subject, and thereby poses the possibility and the problematic of possession, desire, access or use. In an institutional logic, object and subject are mutually bound together through practices. It is through the practices they afford that their objectness is aligned with our objectives. These affordances, the practices and their effects, delimit the objectivity of objects (Boxenbaum et al. 2018).

Second, objectness depends on a social process of objectification. As Latour argues, things can have multiple deployments and their associated ontologies; objects, on the other hand, are those "things" that achieve a stable unification by the "collective" as they are assembled in situations (Latour 2005, p. 115–117, 119). Objectification is achieved in purposive practice. Objectification entails the object's ontological constitution, how it is understood, adopted and used as processes of social construction. This constitution must include the ways in which attributes of objects shape the social construction process, something marginalized in constructivist studies of technology (Leonardi 2013, Leonardi and Barley 2010, p. 32).

Third, an object exists only through its effects in social practice, its product and productivity. Objects are media through which one does things, the reference, signs, sites and most importantly for our essay, media of production. Objects not only afford, they intend and elicit genres of action. From an institutional viewpoint, one of the most important effects of objects is

[9] We obviously do not cover the multiplicity and complexity of the different types of objects (see Nicolini et al. 2012 for an example). The essay focuses on institutional objects only.
[10] One can conceptualize, as feminists have done, the body as the object of the human, that which appears to stand outside our human personhood. For further insights on the bodily experience attached to institutional maintenance, see Stowell and Warren (2018).

the ways in which the practices they afford produce or instantiate particular kinds of goods. This is, as Muniesa refers to simulacra, their "positive reality" (Muniesa 2014, p. 22). It is through these practices that objects become objective in that they enact the objectives immanent to them.[11]

What is an Institutional Object?

Although institutional logics center on material practice, not on positions in fields or persons holding values or schema in their heads, hearts or someplace in their bodily persons, the materiality of that practice has mostly been a vague gesture (Hasselbladh and Kallinikos 2000, Mutch 2018). Compared to Bourdieu (2000)'s habitus as largely non-discursive cognitive capacities and dispositions based upon practical experience of a material habitat, Latour (2005)'s actor-network approach, in which objects are afforded an agential status in conjointly inter-objective and intersubjective social networks, or Boltanski and Thévenot (2006)'s orders of worth approach in which objects equip situations for justification according to common goods via tests of evaluation, the object in an institutional logical approach has been poorly developed, either absent or mute.

In the rest of the essay, we examine institutional objects. Objects are objectified in use; we know them through the ways that we use them. Objects and practices are typically co-constitutive in social life (Heidegger 1962). Institutional objects are good-dependent objects whose objectivity depends on the good they are presumed to produce through the practices they afford, a good whose production depends on the object (cf. Figure 1). The objectivity of an institutional object does not inhere just in its thingness but cannot be separated from that thingness either. Passports, borders and ballistic missiles, for example, are media for the production or enactment of national sovereignty, not just external markers of sovereign authority, but its productive constituents. They are not just its signs; they are its material symbols. There is a particular institutional ontology and teleology built into them.

[11] Our approach of objects through their effects is from this perspective very close to the pragmatist account of the object (Lorino 2018). The concept of valuation is also key to pragmatist authors. Further research could investigate the differences and commonalities between both traditions, potentially envisioning some collaboration, as the one currently ongoing between institutional and practice scholars.

These objects and the practices they afford, and that afford them, come as coherent clusters.[12] Institutional logics are observable constellations of subjects, practices and objects grounded in goods which are done and spoken. They presume a relational ontology and thus requires a relational methodology (Friedland et al. 2020). A relational ontology is required because in an institutional logic subjects, practices and objects are co-constitutive of each other, a co-constitution manifest in their coherence, a coherence that depends on particular substantive goods, goods that participants mark by name. The analytic objects are observable, intersubjectively and categorically marked totalities whose reality inheres in the co-constitutive relations between their elements, not on the net effects of variables as attributes of entities – individuals, organizations, or groups. Institutional logics are complex duality structures (Breiger and Mohr 2004, Mohr and Duquenne 1997). Schatzki, the practice theorist, refers to them as "bundles of practices and material arrangements" or "co-contexturalization of practices and material arrangements," (Schatzki 2002, p. 80–81). Boltanski and Thévenot, the fathers of the conventions of worth school, as "coherent engagement of objects in acts." (2006, p. 148) The same with money, property and market value, ballots, voting booths and democratic representation, borders and national sovereignty, reproductive organs, infants and kinship. In contrast, most objects – like hammers, paper-clips, marijuana buds, and drinking cups – have no internal relation to institutional goods. At the level of production, they are considered technically; at the level of territorialization, it is an issue of social regulation and construction, of which institutional logic will be able to or want to make an exclusive claim to them. Neither their objectivity nor the good depend on the other.

Market devices are institutional objects. Capital, the primary institutional object of capitalism, is a pertinent example. Capital both is and has a value with an internal relation to the good of market value. Muniesa, for example, elaborates a pragmatist formulation of the financial valuation of a firm, following John Dewey in which value is both a subjective prizing and an objective appraisal, a joining of an idea of something to be obtained and something to be done, a practice (Muniesa 2011, p. 25–26). Here the "something" to be obtained is profit, valuation being the capitalization of earnings, a performative and "virtual" operation that creates an asset from an expected future stream of earnings, an operation coincident with ongoing investment and

[12] The genesis of the co-constitution of practices and such institutional objects is an important empirical question whose answer is beyond the scope of this essay. Comparative studies of institutionalization and de-institutionalization are required to answer this question.

disinvestment in the firm in the capital market (Muniesa 2011, p. 30). Valuation is a performative practice, or as he says, "considering a reality while provoking it" (Muniesa 2011, p. 32). In Muniesa's casting value, here an actual or expected market price, is "something to be obtained" through a relation to a "thing," here the corporate securities of another "thing," or "object" or "instrument," here an organization "made fit for valuation" – a corporation (Muniesa 2011, p. 31–32). Akin to Boltanksi and Thévenot (2006)'s orders of worth approach, Muniesa (2011, 2014)'s approach, in fact, brackets the production of market value. Valuation for him is the practice of market evaluation, as manifested in the prices at which corporate equities trade, there being a "correspondence between the act of valuation and the act of purchase" (Muniesa 2011, p. 31). Yet capital, equity, corporation and, of course money, are not "things" at all, nor is the profit for which the organization has been prepared simply a "purpose." These are interlocked institutional objects, a constellation, that depend on the unobservables of market value and property, doubly objective, both goods and reals. The material practices of an institutional logic are not just, as Muniesa (2011, p. 32) translates valuation, "significations," processes by which "something can hold as the sign (read "the value") of something;" they are symbolizations in which values are rendered immanent through institutional objects.[13]

Market value is not just a subjective valuation, it presumes an ontology, market value as an objective fact. This use of the object term is not a materialist retreat, as though material objects ground value; it is that market value is treated as if it were, or could be, a material fact, not as a perspective, an attribute or an arbitrary social construction which drapes the object in an imagined value (Beckert 2013, 2016). Rather the practices accomplished through institutional objects magically produce or carry a value in which participants believe as real and good even before the practices are actualized or after they have failed. Neither norms, nor beliefs themselves secure the value. The value is understood to inhere in the object, to call for particular practices which are understood as the doing of the good. An institutional logic, in its objectivity and perduring coherence, its regularity, covers up and accomplishes the conversion of the made up into the really real; they both reveal and conceal the value that stands forever as a secret hidden in its thingness.

Institutional objects function as representations and representatives, as signs and instances. This is why we use the polysemic term: good, which both denotes a value and a thing. The good

[13] Muniesa has pointed out to us his use of Pierce's understanding of symbol and icon and the way our formulations might align. We have not had time to explore this further (personal communication).

is materialized through practices performed in and through institutional objects. Neither a materialist nor a linguistic account is therefore sufficient to account for the role of institutional objects. An approach that aligns materiality and hermeneutics, that makes the material meaningful and meaning material, is required. To be adequate to institutional reality, it also has to bridge the visible and the invisible. Institutional objects are not just material signs of a category, translations or expressions of an idea (Jones et al. 2012), nor markers of a site where a particular good is pertinent (Boltanski 2011, p. 28, see Boltanski and Thévenot 2006, p. 150). They are not just forms of evidence that justify worth (Boltanski and Thévenot 2006, p. 130–131). Nor are they just material agencies that afford and can be yoked to particular social goals (Leonardi 2013, p. 70). Objects may play all of these roles.

Institutional objects are distinctive, sparse and omnipresent: In practice they are performative, productive, material symbols of the good, common goods deployed and sought after in practice. While the uses of most objects can become institutional, in the sense that there are accepted conventions about their making and their use -- pre-rolled, filtered cigarettes, for example, or they are inscribed with different institutional meanings -- of pleasure, manhood or a health risk, they are not themselves institutional objects. Institutional objects are doubly grounded, both ontologically and teleologically. As the Greek etymology of symbol implies, institutional objects allow the "throwing together" of the material and the metaphysical. Institutional objects are symbolic in the sense that they are intentional, as if they act, or call out to us, as if their use incarnates an invisible good or they call us to such a good. The practices through which we approach them and each other are part of the constitution of who we are in what we are doing: They affect and effect us. They call us to a practice through which we also incarnate these goods; they move us and we move towards them. We, too, are in some sense them. They construct the social as much as the social constructs them. Not only are they -- like capital, offices, accounts, sacraments, votes or parliaments -- objects of institutional practice, real and good, but the practical affordances of the object, its uses in practice, are understood to be productive of the good. They are instrumentally effective, not just because of their material affordances, but because of the good understood to inherently follow from their use, a good whose production cannot be easily partitioned between practice and object. The effects of these material practices are not just a matter of belief; everybody does not necessarily have to agree about them. They just are consequences of practices that deploy them; belief is baked into the material practice whose operations do not just signify but symbolize. They are objectives of practice--through which and towards which practice moves.

In our understanding of an institutional logic, institutional objects and practices are co-constitutive through the good that it is their objective basis. Performativity is typically used as model-dependent practice (MacKenzie 2006) or as iterated authoritative practice (Butler 2003, Culler 2000). In an institutional logic, non-phenomenal goods are part of their production functions. Democracy is produced through practices of election through the institutional object of the ballot. Goods must be believed to be real and good to produce them. Not unlike a mathematical model, they are axioms, perhaps non-logical axioms, of its production of institutional truths.[14] The definition of the situation does not float free. Material practices make regionalized universes of which their logics are constitutive, universes which make their immanent and articulated axioms true. Institutional objects and the practices they afford, the way in which they are used, under which conditions and with which consequences, materialize the non-phenomenal good. They do not depend so much on agreement or rules, as understanding granted by and granting the institutional logic such that the good both appears to and actually does inhere in its operations. As a result, they are "taken-for-granted." Typically, we do not have to agree that something, a piece of land or a song, is property. Property is bought and sold, its value revealed, its teleo-ontology immanent in this exchange process. Like Wittgenstein's understanding of words, the meaning of institutional objects is located in the way they are used in practice, practices grounded in non-phenomenal goods, goods that inhere in something akin to what he termed a "form of life." (Wittgenstein 1953). Belief inheres in the presumptions of practice, in the institutional logic itself. One does not necessarily have to "agree"; they just are.

Objects and Institutional Logics

The Role of Objects in the Institutional Logical Literature

Institutional logical research has been largely immaterial. There are those who argue that the institutional logical approach should eschew the material component altogether. Haveman and

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[14] We are grateful to Caleb Bernacchio for putting institutional logics into relation with mathematics.

Gualtieri (2017), reviewing the literature, admonish: "…definitions of logics as both cultural and material phenomena are problematic. If we are to take the terms we use seriously, we have to admit that logics are cognitive constructs – socially constructed schematas, shared understandings, (preconscious, subconscious, or conscious) rationalizations. Logics are not material constructs: they are not organizational structures, practices, or policies, nor are they rituals or roles" (Haveman and Gualtieri 2017, p. 22).[15] Others have taken critical notice of this lack or silence around materiality (Boxenbaum et al. 2018, Daudigeos and Valiorgue 2018, Jones et al. 2013, 2020). Pinch, a progenitor of the social construction of technology, noted the absence of consideration of how things work in neo-institutional theory and the ways in which technological forms – like DOS, for example (Jepperson 1991), may themselves be institutional, part of conceptual and practical "technological frames." Speaking of the institutional logical approach, Pinch noted that "material practices" are "something akin to the ordinary human practices of everyday life. In effect, material practices are equivalent to social practices. […] [B]y and large the materialness of things does not figure in the analysis" (Pinch 2008, p. 464). Pinch is largely correct. When institutional logical theory has broached the role of objects, it has tended to envision them through the rubrics of reference, signs or sites, neither as constituents of practice, media for the production of particular goods, or material symbols of those goods.

Objects as reference. Many institutional logical studies have focused on the object as a question of reference, how different objects are taken up by different institutional logics, the way, for instance, recycling was transformed from a service promoted by activists in the name of community and nature to a profitable commodity promoted by professional technocrats (Lounsbury et al. 2003), American publishing firms, in which the book moved from a literary production in which editors had primacy to a profitable commodity in which managers had control (Thornton 2004), the commodification of scientific university research as patented commercial property (Colyvas and Powell 2006), or how the dominant producers in the French musical industry transposed the meanings associated with CDs to new digital technologies in order to maintain intellectual property rights, despite the difference in these artifacts' physical properties (Blanc and Huault 2014). Other examples can be found in Raviola & Norbäck (2013)'s study of

[15] Mutch (2018), critically pointing to the absence of the organization per se in the institutional logical approach, argues that as "bundles of practices," organizations, and the structure of positions composing them, stabilize and institutionalize them. He also points to the systematic way that organizational forms – like corporations, hospitals and armies – "mediate particular forms of logic" (Mutch 2018, p. 246–247).

how going online in 2007-2008 impacted the journalistic practices of the largest Italian financial newspaper by transforming the print newspaper into a "law book" that gave meaning to the digital version.

Objects as signs. Other studies have pointed to the ways in which the material specificity of practice is integral to the formation and fixing of institutional logics, giving primacy to objects as "signs." For instance, Jones, Maoret, Massa and Svejenova (2012) study how two genres of the new category of modern architecture – organic and functional – "enacted" distinct institutional logics through different "artefact codes." The authors showed for instance how the way architects combined stone, brick, concrete, steel and glass to distinguish their production from the dominant historicism of the profession and from each other. In their formulation, the code rules of inclusion and combination "expressed" the category, while institutional logics provided the "content" – assumptions and values – that motivated and legitimated the artefact code, as well as guiding the "theorization and specification of artefacts' features, forms and functions." (Jones et al. 2012, p. 1524–1525). Rao, Monin and Durand (2003) describe how culinary professionals in pursuit of creative autonomy motivated by the anti-schools spawned by May, 1968 created nouvelle cuisine as an "identity movement" through new "framings" based on the "organizing principles" of a new institutional logic organized around a new "truth" of food, with new role identities for the chefs who promoted and adopted it. Nouvelle cuisine entailed not only the promotion of the autonomy of the chef and the erosion of the power of the restaurateur. It involved a material transformation of the "grammar of culinary practice," including short menus, low inventories, fresh ingredients, and service through the plate, as opposed to the waiter's rites outside of it (Rao et al. 2003, p. 798, 800).

Objects as sites. A few studies have described materiality in terms of physical spaces in which particular forms of practice take place (Raaijmakers et al. 2018). At the most rudimentary level, a site is a container or location of practice, whose attributes enable or constrain certain forms of practice. For instance, De Vaujany and Vaast (2013) demonstrate that organizational spaces constrain and enable the unfolding of new institutional logics, through a study of the transformation of the former NATO headquarters into a university (see also Lanzara and Patriotta 2007, Lawrence and Dover 2015). In these cases, the site is a context external to the practices that unfold within it; it is a location.

Objects and the practices to which they are joined come in congeries, as in Heidegger's conception of equipment (Heidegger 1962). Mol remarks, "Even when it is not being interpreted, matter is never alone. For it may well be that matter acts, but what it is able to do inevitably depends on adjacent matter that it may do something with" (Mol 2013, p. 380). Objects thus function as constituents of a social order in Schatzki's sense of "arrangements" "a hanging together of entities in which they relate, occupy positions, and enjoy meaning" (Heidegger 1962, Schatzki 2002, p. 20). Where objects are an inherent part of this context they are part of the constitution of "sites" of practice, practice being sets of "temporally unfolding" nexuses of doings and sayings (Schatzki 2002, p. 65, 72).

Jerusalem's Western Wall certainly qualifies as a site of practice. Using Lefebvre's tripartite understanding of space to analytically stratify the multiple meanings of Jerusalem's Western Wall, Preminger and Drori (2016) study the ways in which competing institutional logics – religious, archeological and nationalist – are differently, but coherently, "spatialized" in the planning conceptions, lived practice, and linguistically expressed perceptions of this site (see also Drori and Preminger 2018). Here the institutional ontology of the wall is continuously at stake. Different segments of the wall have been allocated and claimed by different authorities and their clienteles, grounding their claims and practices in different goods – the central sacred site of divine presence by religious Jews, the layered remains of civilizational heritage by professional archeologists and tour-guides, the site and warrant for the founding of Israel as a Jewish nation-state by Zionist political elites and social movements. Each has distinctive practices by which it relates to the material site, from tactile prayer to corporeal distance and access through monetary payment, for example. We could read the relationship between the multiple institutional logics and this singular stone wall as a question of reference, as a practical and political contest between communities with divergent cultural constructions over an external material site, not unlike whether recycling would be a community good or a commodity (Lounsbury 2001, Lounsbury et al. 2003). But that would not be adequate here, at least for the religion and the nation-state. This retaining wall is a material symbol of the good that is "enacted," the site is a productive symbol of the good which makes it a site in the first place. It is, for instance, because this retaining wall bounds the ancient center of divine presence, both a historical center of sacrificial pilgrimage and ancient nation-state formation, that both locates and grounds the origins of both Judaism and Zionism, that has afforded the migration of prayer into every segment of the Wall and the "porousness" of the nationalist logic, noted by the authors. For both Judaism and Zionism, the Western Wall is what we would term an institutional object, which we explain below.

A similar account of the role of institutional objects as sites can be found in Jones, Lee and Lee (2020)'s study of Boston's North End neighborhood. The authors show how the institutions of religion, state, and ethnic groups were materialized in the place through the people, notably the ethnic communities – what they refer to as "embodied materiality" – and the buildings of the neighborhood, a materialization that proved essential to the institutionalization process itself. Also mobilizing the concept of place, Rodner, Roulet, Kerrigan and Vom Lehn (2019) uncover how the institutional work over the past twenty years in Venezuala's art world unfolded through the physical use of the territory and its buildings, such as galleries, museums or the urban planning of the city of Caracas itself. Despite this work, the processes through which the material affordances of the places themselves shape the institutional practices are little understood, calling for further work (Jones et al. 2020).

The Institutional Productivity of Objects

In the institutional logical literature, objects have tended to be theorized as material signs or expressions of a logic or as domains of exterior objects over which divergent institutional logics contest for hegemony, neither as good-dependent in their objectivity, nor as media through which particular logics are constituted nor institutional goods produced. Acknowledging the meaningful materiality of institutional objects offers the institutional logical approach one pathway towards those concerns. Nevertheless, several scholars have pointed to the productive role of objects in the unfolding of institutional logics.

Good-Independent Objects. Some emphasize objects as good-independent instruments. In his historical genealogy of the material practices and devices of modern rationality, Quattrone displaces a belief-based substantive rationality in favor of a material procedure unhinged from it. Quattrone traces the modern form of accounting as visual techniques of knowledge organization to late medieval and humanist rhetorical practice he identifies in both the spiritual exercises and the administrative accounting procedures of the Society of Jesus, founded in the 16th century by Ignatius of Loyola (Quattrone 2015). Quattrone shows how Jesuit rationality was grounded in materially specific practices, not in their variant of Christian belief (Quattrone 2015, p. 436). Jesuit spiritual exercises appropriated humanism's rhetorical material practices – its mnemonic ordering, its analytic composition of an inventory through "acting images," *rationes* – such as diagrams, maps or wheels, as a machinery for the production of thoughts about God. For the Jesuits, it was not a structured inventory of propositions, but of sins to be visually, corporeally and affectively remembered, examined and extirpated, a necessarily incomplete practice driven by the *intentio* for God, a practice driven by a desire to perfect an imperfectible self for the sake of the not only the invisible, but the inaccessible God. These were not representations as much as practical pathways, marked by images and intense feelings, not ideas, but sights, tastes and smells, sometimes modeled on and as liturgical pathways, likewise journeys towards heaven or hell, an orthopraxis adopted by the Jesuits to lead their followers not to a cognitive idea, a good argument, or a "factual representation," but to illumination, to *theoria*, "a vision of God," an aching desire for impossible visibility. In this "unfolding rationality" of necessarily incomplete representation, the Jesuit order "operated as a gigantic rhetorical machine, which made the continuous reinvention of its reason of existence one of its main features" (Quattrone 2015, p. 435). In this rendering, the material means were themselves meaningless, except as means of making meaning.

Good-Dependent Objects. For others, objects are understood to have an internal relation to the ontology and teleology of the goods produced. In his comparative history of cotton mills in England and Germany, Biernacki (1995, p. 143) writes of "the lucid ideas of a culture that are incarnated in material techniques." He demonstrates the materialization of different ontologies of labor – labor as a commodity owned by the English worker appropriated through its product, the quantity of which as the number of "picks," determined the worker's pay, versus German labor power appropriated in and as labor time, paid by the quantity of its activity, the number of shuttle movements, or completed "shots." These ontologies were not only integral to ways in which workers organized and the dominant conceptions of political economy, they were materialized in the organization of their factories: the English built a hard boundary between inside and outside with a single entry for the workers, the point of exchange, whereas the German factories had more porous boundaries with the outside, but physically partitioned the work process so as to enforce the attachment of the worker to their machines, the point of production, to facilitate observation and prevent the intra-factory worker movement.

Others point towards a co-constitutive relationship between goods and objects which do not specify the primacy of either. Mohr and Duquenne (1997) use Gallois lattices to study the structural dualities of category and practice among relief organizations in New York City at the turn of the century. They demonstrate how material practices – the provision of money, food, shelter, or work – were applied to particular categories of people without money, the "working poor," the "distressed," the "destitute," and the "needy," while reciprocally the categories were known and enacted through the distinctiveness of their practices. Categorical meanings inhere not only in the association and differentiation of linguistic terms, but in their practical entailments, and by implication, vice versa. Although they do not develop this, these couplets of practice and category are grounded in different ontologies of "poverty," in its sources and nature, as the voluminous and contested discourse attending their emergence attests. Whether certain attributes of these material practices afford these ontologies and/or are afforded by them however has not yet been addressed.

For Mol there is no singular object; there are different versions of it, performed in specific practice. She writes of the way that particular practices enact "objects that carry new realities, new ontologies with them" (Mol 1999, p. 75). She does not invoke the performative term. In her studies of anemia, for example, Mol points to the ways in which different diagnostic practices -- clinical, laboratory, and pathophysiology -- "enact" or "perform" different versions, that is different ontologies, of the disease -- visible symptoms and fatigue, low hemoglobin counts based on statistical distributions, the individualized hemoglobin level necessary to transport oxygen through the body (Mol 1999, p. 77–79, 2002). Although she does not thematize it, these ontologies and the practices that "perform" them are all grounded in a single teleology: the health of a human body which is not identical to any of them. The performance of the ontologies is grounded in this deeper onto-teleology.

Institutional Logics and the Sociology of Valuation

Market Devices as Institutional Objects

The sociology of valuation largely inspired by the work by Boltanski and Thévenot (2006) explores the mechanisms through which valuation – what is deemed of value – and evaluation – how to assess that value – is agreed upon, particularly in markets (Helgesson and Muniesa 2013, Vatin 2013). Market devices are essential to this process. A market device is a simple way of referring to the material and discursive assemblages that intervene in the construction of markets (Callon et al. 2007). Such devices include financial models, or accounting frameworks that "aim at rendering things more "economic" or, more precisely, at enacting particular versions of what it is to be "economic'" (Callon et al. 2007, p. 4). This body of work argues that market devices can "perform" the reality they embody (Ferraro et al. 2005).

The dominant understanding of performativity is a theorization effect, not as a legitimation, but as a self-fulfilling production-function (Gond et al. 2016, Marti and Gond 2018).[16] Its classic example is the historical analysis of the emergence and decline of the Black-Scholes-Merton model of options pricing at the Chicago Board Options Exchange by Donald MacKenzie and Yuval Millo (2003), about which we will have more to say below. The authors show that the more financial actors used the Black-Scholes-Merton model to price options, the more options were priced according to the model. For them, it is a picture book example of "Barnesian performativity" of economic theory, the way in which the economy comes to conform to an economic model through what the sociologist Barry Barnes saw as "self-validating feedback loops" (MacKenzie 2006, p. 19, MacKenzie and Millo 2003). These "loops" are not just actors' expectations or beliefs; they refer to the ways in which the economic model is "incorporated" into the "infrastructures of markets:" its "algorithms, procedures, routines, and material devices" (MacKenzie 2006, p. 19). Barnesian performativity is where abstract representations and symbolic languages are incorporated into effective material practices. It depends on three contingent links: material incorporation into objects, elaboration of practices using those objects, and effects of those material practices consistent with the representation. Other similar examples can be found in other settings, such as carbon markets (MacKenzie 2009) or the creation of financial value out of the use of environmental, social and governance (ESG) criteria (Arjaliès and Bansal 2018, Marti and Gond 2018).

Market devices are institutional objects whose efficacy depends on the good of market value, the classic locus of the apparently ontologically objective material in the market as opposed

[16] There are different theoretical accounts of performativity (see Garud and Gehman (2019) for an overview of recent debates). Gond, Cabantous, Harding and Learmonth (2016) identify five conceptions of performativity used in organization theory: doing things with words (Austin); searching for efficiency (Lyotard); constituting the self (Butler, Derrida); bringing theory into being (Callon, MacKenzie) and sociomateriality mattering (Barad). This essay does not pretend to give an exhaustive account of this process; it solely focuses on institutional objects and its four value moments.

to the ontologically subjective ideal world outside of it.[17] Market value is an unobservable good realized through variable practices by which quantities of priced commodities are exchanged between buyers and sellers. The institutional logic of the market is located primordially in the material practices of priced exchange, neither in the utilities derived from the use of variably scarce commodities exchanged, nor the abstract labor time embodied in them (Orléan 2014). Although there is no market value without a money price, money's market value is problematic in economic theory. In the deterministic general economic model, each good has a utility while money as such has none except as a liquidity preference. It is only in a stochastic world that money can have a sense as a hedge against uncertainty (Maurer 2002).[18] Economics cannot easily theorize money's institutional quality. Dodd (2016, p. 8) – a sociologist of money, for whom money should be envisioned as a "social form" explains: "economists define money according to its basic functions; or, following Keynes, they distinguish between money's abstract role as a *money of account* versus its properties as a *medium of exchange*." Yet money cannot be reduced to a medium of exchange, nor can itself be separated from private property (Yuran 2014). "[M]oney can be seen as the pinnacle of private property," the economic philosopher Yuran writes "What it carries to an extreme is precisely this element of exclusion, of *not yours*, that is essential to private property. Money is the extreme form of ownership because it has no other quality but ownership…. Yet, precisely as an extreme form of private property, it has nothing truly private in it: it is completely meaningless as a private object" (Yuran 2014, p. 76, 138). The same goes for property. While economists recognize that property is a precondition of exchange and, since Coase, that the nature of property rights shapes allocative efficiency, they do not endogenize property in their models. What property is and what is property is not derivable from iterated self-interested individual interactions: It, like money, is an institutional object.

Institutional Conditions of Valuation

The institutional dimension of the performativity of market devices is not inherent. Beunza and Ferraro (2019) propose a model that bridges performativity and institutional theory, using

[17] Note that critical accountants have long questioned the external objective account of market value (see Miller and Power (2013) for a review).
[18] Conversation with Daniel Gabay, February 1, 2018, Paris.

responsible investment as their empirical context. They suggest that a market device cannot "perform" the reality it presumes if actors do not conduct some institutional work, understood as the "purposive action of individuals and organizations aimed at creating, maintaining and disrupting institutions" (Lawrence and Suddaby 2006, p. 215). "Performative work," the authors argue, precedes performativity. Performative work, it appears, involves such things as translation – understood as the construction of a new socio-technical agencements (Callon et al. 2007, Callon and Muniesa 2005) and enrolling executives and others to set up the appropriate institutional conditions for the device to be adopted.[19] For that institutional work to work, particular institutional constellations must already be in place. For example, in her analysis of the different valuations of nature in the wake of massive tanker oil spills in France and the United States, those in France much lower relative to the damage, Fourcade (Fourcade 2011a) explains:

> I show that the mere availability of certain economic technologies does not guarantee their performative effects for the simple reasons that these technologies may not muster enough institutional and political support or that they may not resonate enough with the cultural claims they are supposed to represent. In fact, economic methods for the valuation of nature come in very different shapes and imply very different forms of calculability. Neither of these are incidental, of course; rather, both the methods themselves and the calculability they embody are the product of very specific social processes that are of great relevance to the 'performed' outcome itself. (Fourcade 2011a, p. 1724–1725)

The implication of Fourcade (2011a)'s approach is that "economic technologies" mediate dominant institutional logics, that the capacity for, the nature of and the legitimacy of market valuations is contingent upon the institutional formations in which they take place. Institutional formations, in her words, "carry" "economic valuation techniques" (Fourcade 2011a, p. 1726, see

[19] The authors show that the customers of Visuals – a data provider on financial markets – adopted the ESG (Environmental-Social-Governance) product that they first refused. However, the authors do not explore whether the good of market value was produced and apprehended differently following this adoption. Nor do they specify which institutional changes made those changes possible, probable or necessary. Whether social and environmental responsibility became part of the pricing of assets through the ESG criteria remains to be explored. Likewise, Marti and Gond (2018) propose a framework that comprises six boundary conditions for theories to become self-fulfilling. The authors identify the transformation of "social norms" following the use of material devices by high-status actors as the first key mechanism through which theories perform the reality they envision. Yet, the authors do not elaborate on the mechanisms through which practices shift through the use of material devices, nor analyze their relationships with the social norms they point to.

also MacKenzie et al. 2012, p. 292). The models incorporated into market devices in this case have effects because they are already built around and consonant with the dominant institutional ontologies. The Americans' choice of passive use or lost use valuation techniques, also known as contingent valuation, to estimate non-market, non-use losses – species and tidelands that neither enter the market nor are used by anybody in particular – is revelatory of the institutional basis of a market device, and its ability to value the material world according to a market logic. Given that there is no effective demand, contingent valuation seeks to simulate an imaginary market using survey questions about how much people would be willing to pay to avoid or obtain a particular cost or benefit – a fictional expectation (Beckert 2016). This valuation of a public good was grounded in the aggregation of individual preferences expressed in monetary terms. The survey was not even administered to Alaskan residents. Such a technique was neither thinkable nor actionable in the French case because animal species per se were not property, because the referential scope of the market was both restrictive and suspect, because state authority is grounded in its defense of common goods, and because the public good, and the social goods provided in its name, is neither an aggregation of citizen preferences, nor more pertinently, contingent on its price, nor citizens' tax contributions. In point of fact, the French, who were most hostile to the intrusion of the market in the public sphere, held to a more restrictive understanding of the market, whose valuation techniques were only applicable where things were commodified, thus rejecting market estimations of the value of lost biomass based on the closest marketized equivalents and even refusing the property claims of local townships to non-commercialized biomass (Fourcade 2011a, p. 1755–1756). In America, unused nature was the property of all citizens; in France unused nature could not be property – of anybody.

The Institutionalization of Goods through Institutional Objects: The Four Value Moments

Good: The Fusion of Fact and Value through Institutional Objects

We aim to model some of the primary mechanisms through which institutional objects, such as market devices, produce goods, such as market value. To do so, we need to restore value as a good which is both ontological and teleological, a good which is constitutive of and constituted by the central objects around which the practices of an institutional logic are organized. In our conception of an institutional logic, values are objectified in material practices, not just as normative deductions, nor subjective ends, but as objectivities immanent in productive practices. For institutional objects, there is, as Barad contends through her concept of intra-action between subject, apparatus and object, an internal relation between the good, practice and object (Barad 2003). One only knows the goods as effects of the core practices of an institutional logic, practices which depend both for their existence and their effects on institutional objects that objectify those goods, and hence on belief in those goods and their internal relation to the objects in practice.

Value has been progressively orphaned by one social science discipline after another, a conceptual ancestor who no longer amounts to much. As economists abandoned value, whether use or labor value, for preference and utility functions (Martin and Lembo 2016, p. 9), sociologists turned to the value term as an explanation for individual actions. Value was largely located in individual subjectivity as a commitment to abstract ideals, an exogenous preference, a reason for action, or as a normative model of how one should ideally conduct one's life (Khaire and Wadhwani 2010, Weber et al. 2008). If values are "internalized reasons" for action, that individuals' behaviors do not accord with their values and that they are demonstrably incapable of giving "coherent accounts of the reasons why they engage in certain lines of action" has called into question the role of values, or any internal representations, as causes of that action (Lizardo 2017, Lizardo and Strand 2010, p. 205–206, Vaisey 2009) .

Other scholars shifted from value as a substantive end, a subjective commitment, or a propositional discourse shaping action to a justification for behaviors dictated by, but not guaranteed or constituted by an exterior institutional structure. In the "toolkit" approach, for example, culture shapes action not as value as internalized end, but by providing a repertoire of actionable means (Swidler 1986, p. 284). Swidler argues that culture is elaborated to solve "problems of action" where institutions are incomplete, romantic love, for example, emerging to manage the decision to mate and marry, to enter an external structure of lifelong monogamy under conditions of free individual choice (Swidler 2001, p. 156–158). Love is a members' orientation to the institution of marriage, not a constituent of it (Swidler 2001, p. 176). Institutions are external structures which support behavioral regularities without assuming much individual cognitive (let alone coherent) representation (Lizardo and Strand 2010, p. 206–207). The "myth of love" is, as Lizardo and Strand (2010, p. 214) put it, "descriptively sterile but pragmatically useful."

Discourses of value – like love – appear at the limits of externally objectified institutions, not as their animating core.

Martin and Lembo (2016) suggest that we "banish" the category of value. The proper grounding for a "phenomenologically plausible" account of action, they argue, is interest, as a quality of an individual's engaged and invested relation to external objects into which he projects himself. "We see no reason," they write, "why we should posit "values." But when it comes to interests, the objects are already there. The interests are no tertium quid, they are the subject's relation to the objects" (Martin and Lembo 2016, p. 25). Values, they argue, are claims about the validity of one's interests, "virtual entities" deployed in particular where there are no norms or norms are disjoint (see also Martin 2011, p. 253). Martin's epiphenomenal subordination of value follows on Bourdieu's field theory. Goods appear in Bourdieu's approach as an *illusio*, the value of "stakes" in an "enchanted relation to a game," "stakes created in and through the fact of playing," not as goods, values and ontologies that ground observable social relations (Bourdieu 1998, p. 77). The social relations in a field have primacy, not the good. The *illusio*, and the investment of which it is both producer and product, is an intermediary, derivable from the consonance of a structure of positions and dispositions that order agonistic relations of domination based on distributions of different species of capital (Bourdieu 1998, p. 47, Friedland 2009).

We would argue that the capitals, and by implication the structures of positions, upon which field theory focuses are likewise grounded in these goods, immanent in their objectifications and thus the practices these objects afford. Objects do refract social relations, our relation to others apparently inhering in their qualities, our sense of value an expression of our agonistic social relationships in specific fields (Martin 2011, p. 227–228, 252–243), but so too those social relations refract the non-phenomenal goods immanent in actual objects, affording the practical relationships we have with the objects and thus each other. The practical oughtness of institutional objects is good-dependent. The goods immanent to those objects and the practices they afford ground relations as much as they are their refractions.[20]

Institutional objects both afford and signify forms of practice that constitute the objectivity of the world, as a world that is, can be and ought to be organized around particular goods. Institutional objects are vehicles by which goods are instituted, produced, evaluated and

[20] Likewise, the pragmatist approach seems to emphasize the productivity of relations of signification as interactive processes of action, occluding their potential good-dependence (Muniesa 2007).

territorialized.[21] In the case of market value for instance, market devices and the models of the world which they incorporate are not simply representations of the real, they become objectified ideals, practical norms and automatic decision-rules and routines by which to evaluate and act in this world, thereby producing a world so modeled (MacKenzie, 2006, 2011). Through this process, the material cartography of an institutional logic is re-built or fashioned anew. Not only is it through new reference that ontologies of existing objects shift, but the introduction of new institutional objects can transform institutional logics themselves.

The problem in thinking and researching this is that the variably constitutive role of objects to institutional practices and their co-constitutive relation to institutional logics has barely been studied (see above). This weakness is not confined to the institutional approach. The orders of worth approach (Boltanski and Thévenot 2006) recounts how conflicts and controversies about worth unfold, but now about how a new order of worth is instituted and equipped, formatted with new objects. Given the centrality of objects in qualifying a situation, the approach is necessarily weak on the moment of instituting, manifest in its dependence on what are assumed to be trans-situational philosophical principles of centuries past – those of Augustine, Hobbes, Rousseau, for example – to distill the "grammars of the political bond" (Boltanski and Thévenot 2006, p. 65–67). The implication is that historical variations in the equipment are not substantially important. In their account, it is members' "reason," not their practical relation to objects, that draws them back over and over to the same principles. In Latour's emphasis of inter-objectivity, as complement, if not endoskeleton, to inter-subjectivity, many objects don't really do anything, there as what Latour distinguished as intermediaries as opposed to mediators that "*make* others do unexpected things" (Latour 2005, p. 106–107). Nor has the relationship between what Leonardi (2013) calls the material agency of objects, as opposed to their social agency, been properly specified. The constitutive role of objects in an institutional logic, their role in the emergence and institutionalization of one logic as opposed to another, or how the ontological and value entailments of a logic afford the fabrication or appropriation of particular objects, has not been systematically addressed (Leonardi and Barley 2010).

[21] Or as has been argued about accounting, both a "means of representation and means of intervention." (Kurunmäki and Miller 2011, p. 222).

Institutionalization of the Good: The Four Value Moments

Non-phenomenal goods are institutional, not as task-irrelevant cultural constraints on a technical core of activities that must be buffered, nor a value or category imposed on an ontologically objective objectivity, but as a constitutive component of some of the most important practices in which we regularly engage and the objects with and through which we execute them. Such institutional goods ground a lot of our most important social facts, the worlds we take for granted and care most about.

We argue that the objectification of a good operates through four value moments: institution, production, evaluation and territorialization. These value moments are not independent of each other and should not – although we number them below – be viewed as sequential, but as co-existing and co-implicated practices. Each good depends on all of these moments to exist as both fact and value.[22] Each of the value moments is potentially subject to contest, including political contest. Because non-phenomenal goods ground these value moments, belief is central to the institution, not as creed, but as the ground of practice, as the understanding that makes a practice what it is (see Dromi 2019). Belief affords the means for critique and conflict, restoring politics to the study of institutional logics. Belief in a good depends on its actionability in the world: how it is produced, how it is measured, and to what it can refer. Reciprocally, each of these moments – institution, production, evaluation and territorialization – depends on an underlying belief in the good. Each of these moments operates through an objectification, both institutional objects that have an internal relation to the good, and non-institutional objects that do not.

Each moment is a kind of institutional practice – both ordinary and extraordinary, whether every day, irregularly recurrent, or original and singular. Evaluation can fail; territorialization can be contested, production functions break down, be modified or transformed; and the credence and faith in the good necessary to institution can erode, be passionately and violently contested, and sometimes reveal themselves to be fatal illusions. Because social fields are institutionally complex, situations composed of multiple institutional logics a matter of course, and because people can and do make recourse to alternative goods, the institutional moment is an omnipresent possibility (Boltanski and Thévenot 2006, p. 151, 216, Pache and Santos 2010, Smets et al. 2015, p. 289).

[22] We have used the temporal category of moment to highlight the processual aspect of practice and the animated movement implied by an institutional logic.

Each moment is, whether implicit or purposive, a kind of institutional work; together they are what make institutions work (Lawrence and Suddaby 2006). The nature, co-dependence and coherence of these value moments constitute the grammar of practice of an institutional logic. Our theoretical model is summarized in Figure 2.

Insert Figure 2 about here

1st Moment: Institution. The moment of institution is the establishment of a good, a belief that a particular good is both valuable and realizable, both ideal and real. People have to believe that the good is an actionable idea immanent in particular forms of objectified practices that produce the good they presume. They have to imagine that the good can exist and be produced through the institutional objects.[23] In the moment of institution, fact and value are fused through the fashioning of institutional objects and the practices they afford as opposed to the Weberian opposition of instrumental and value rationality, based respectively on the lawfulness of ontologically objective objects and the willfulness of subjective subjects (Weber 1958, Weber et al. 2008). Institutional practices are enacted through and upon objects, objects which are either constitutive of those practices, institutional objects like commodities, or are appropriated by those practices, as their legitimate referents and/or domains of operation, non-institutional objects like a particular product or service, a cheese or recycling. This, of course, points to institutional logics as constellations of material practices, as networks of objects and practices, something beyond the scope of this essay. Institutional objects are not only ontologically objective. Institutional goods are ontologically subjective as well; and although they are understood to be immanent in particular forms of material practice, they can never be reduced to them. Just as market value exceeds, but cannot be separated from, the transfer of ownership of a good occurring at a particular price, so romantic love exceeds physical intimacies, cohabitation and performative speech acts for which it is a condition of possibility.[24] Both market value and romantic love are goods whose existence cannot be derived either from the senses nor from reason, but whose presumed existence is

[23] Theorizing the antecedents of the four value moments needs further research. This essay does not investigate the processes that lead to some institutional logics and/or institutional objects to be selected. Recent work investigating the role of imagination and fictional expectations in the production of social life (Augustine et al. 2019, Beckert 2013, 2016) offer useful insights into those mechanisms.
[24] This is true in markets as well: All information is not reflected in price; uncertainties abound. Financial assets are assumed to proliferate to price these uncertainties..

necessary to buying and loving. Each depends, as Weber claimed about any religion, on faith, on a sacrifice of the intellect (M. Weber, 1958). Although it may not be declared or spoken, belief always lurks in moments of institution.

In an institutional logic, value is not just subjectively added as an exterior valuation of an ontologically objective thing. Nor are practical relations to objects just rational responses to laws of cause and effect of ontologically objective things. The good becomes immanent to both the object through which and the objective towards which it is deployed in practice. Desire is thus a necessary constituent of a good: The good is something to be wanted and pursued. The taken-for-grantedness that is the hallmark of institutional logics depends on belief as affective credence, in knowing and wanting, in the fact of this value and the value of this fact. Subjectification and objectification are manifestly co-dependent in the moment of institution, as the proliferation of "entrepreneurs," leaders and icons of change indicate (Battilana et al. 2009). In the moment of institution, belief matters and is material. Instituting entails objectification. And objectification, as Boltanski and Thévenot have pointed out involves the making of a "world," which is "the coherent engagement of objects in acts" (Boltanski and Thévenot 2006, p. 40, 148, Daudigeos and Valiorgue 2018). This constitution of value is a moment that can neither be fully grounded in the senses nor reason, nor the materialities of its practice. Although institution is most evident in the moment of inauguration or initiation, it is also immanent in the reproduction of material practice, particularly where institutional logics are incomplete, where they fail or are contested, in moments of exception (Derrida et al. 1992). The inaugural or restorative moments of institution require faith in non-phenomenal, incommensurable goods as the ground of practice. It thus affords opportunities for passionate and even violent conflict as a condition of instituting, for the possibility that for want of objects persons themselves will seek to incarnate and emblematize an incommensurable good, "himself as the measure of all things" (Boltanski and Thévenot 2006, p. 229–230). It not only implies belief in the reality of the value and its objectifications, it entails complementary belief in the limited worth or even worthlessness, if not badness or evil, of other institutional objects and the hence the goods upon which they depend (Weber 1958).

Institutional objects are materializations and makers of the good. But they also take on a symbolic quality as its embodiments and instruments, as markers or signals of the good. They are material symbols. They tell others that you are going to do something specific here. And it is as in this latter form – particularly as they appear as sites – that they are often thought. In the orders of worth approach, objects function primarily as formats, “arrangements of objects that qualify the various situations,” part of the *dispositif* that delimit zones of pertinence of different common goods (Boltanski 2011, p. 280, Boltanski and Thévenot 2006, p. 216). So, too, in Schatzki’s theory of practices, which are nexuses of doing and saying, objects are primarily “symbolic markers which establish certain settings for certain practices” (Reckwitz 2002, p. 211, Schatzki 1996, p. 189). When objects are out of place, unconventional and anomalous to a “natural” situation, they offer grounds for critique, the unfolding of controversy and demands for purification. These objects materialize the good, an apparent materiality that grounds the facticity of the value. Institutional objects in these cases are not just signs; they are productive and performative material symbols.

2nd Moment: Production. The second moment is a production function. It consists of fixing the practices and the objects by and through which that good is to be produced or enacted, by which the good is to be made, realized or materialized in the world. The production of a good always entails specific forms of networked and layered material practices.[25] Goods are believable because they are materially actionable, and they are actionable because they are believable. It is in this moment that institutional objects and goods take on a co-constitutive relation, such that objects become part of a symbolic machinery, not arbitrary signs, nor markers, but makers or constituents of the good. Unlike naturalistic approaches to value in economics, goods can be produced by institutional objects only through practices of valuation. It is through these practices that the ontologies of value are objectified in the world. Yet, people experience value as inhering in the objects themselves and the practices through which they are objectified. They experience this “isness,” this transcendent immanence. They act as if “it” exists; they behave as exterior subjects who have relations to objective things. They do not live their institutional lives as social constructivists. This conversion of the relation of value and object into a transcendental-immanent entity, this experience of the thisness of a good, is, akin to Marx’s commodity fetishism, where relations between men become the living principle animating things (Marx 2019). But unlike it,

[25] Remarkably studies of production and their impact on the organization of work, like that of intelligent technologies, tend to focus only on a focal technology, and neglect its relation to the larger technical “stack” – focal application, platform, subsystems, components, among others, and the role system which produce it (Barley and Beane 2020) (Barley and Beane, 2020). In their review of the literature on intelligent technologies, they write: “We have yet to find an analysis of the implications of an intelligent technology for work and employment that traces the ramifications of a focal application vertically through a technical substrate” (Barley and Beane 2020, p. 39, but see Schatzki 2002).

the experience is not a false consciousness, as much as it is a consciousness necessary to the operational effectiveness of institutional objects. Indeed, for example, when the assumptions of monetary policy had been taken-for-granted, the Federal Reserve chair's making them explicit between 1998 and 2014 exposed their contingent constructedness and created uncertainty in the financial markets (Harmon 2019). The taken for granted facticity of institutional objects in a coherent grammar is a belief in an immanent good.

Good-dependent institutional logics, and the institutional objects that anchor them, are modes of production before they are grounds of justification or qualification, unlike the orders of worth perspective (Boltanski and Thévenot 2006). The orders of worth framework emphasize the ways in which objects "objectify the worth of persons involved" (Boltanski and Thévenot 2006, p. 142). Production, as a general term, enters as justification: The "investment formula" that characterizes each order of worth is a doubly faceted set of opportunity costs, both the privation of personal pleasures that have no relationship to that worth and the privatization of other collective goods that can only appear as personal pleasures (Boltanski and Thévenot 2006, p. 142). In an institutional logic, the moment of production captures institutional logics as modes of production, not as, in the orders of worth perspective, as modes of coordination, where practices are understood primarily as establishing "relations of equivalence" between subjects and objects, observable in distributions of worth, "making the worth of persons manifest" (Boltanski and Thévenot 2006, p. 131, 143). In the moment of production, institutional objects produce the good; they do not, as in the orders of worth perspective, primarily "substantiate" worth (Boltanski and Thévenot 2006, p. 131). In this moment, institutional logics are modalities that both produce and are produced by the good. The good and the object have an internal relation mediated by the practice that the object affords. They can neither be reduced to categorization, to exterior social construction, nor to Latour's mediators and intermediaries (Latour 2005). As Latour has recognized, situated actor networks of heterogeneous beings cannot by themselves generate "the value that emanates from that situation," (Latour 2013, p. 42, 43) that defines its activity, provides its continuity, and grounds its mode of veridiction.[26] One only knows the goods as effects of the core practices of an

[26] In his subsequent work, Latour (2013, p. 357–359) explains that law enjoys a unique form of autonomy, since law comprises its own explanation, by virtue of its own formalism. Law is certainly a domain in which institutional objects play an essential role. Beyond law as a setting, further research should investigate the broader role of law in the theorization of the four value moments, notably through its influence on how we value "things" (Muniesa 2015, Thomas 2002). We are grateful to Fabian Muniesa for raising this point.

institutional logic carried through institutional objects, practices which depend both for their existence and their effects on the presumed existence of these non-phenomenal goods. In religion, an account of religious practice based solely on things and words would be a form of idolatry; in social theory it takes the form of proceduralism, promoted by those who believe that our explanations of modalities of action can do without the realm of ends or what Aristotle referred to as "final causes" (Lizardo and Strand 2010, Quattrone 2015, Swidler 1986).

3rd Moment: Territorialization. The third moment is that of reference, to what objects, practices or beings can value be attached, and hence through which it can be produced and evaluated. Matters of institutional fact involve not just what property is, for example, central to the first moment, but what can be property. Is a human body, a body part like blood or a kidney, or a genetic sequence property? What is the referential specificity and range of an institutional logic; to which domains does it refer? What aspects of nature – air and water, for example, can become property rights priced and traded in the marketplace? It is here that ontological politics are concentrated, where controversies over what this/these is/are explode, both in movements of institutional frontiers, in moments of instituting and de-instituting. This is the moment where words and things are self-consciously wedded to power and coercive apparatuses, weaponized as rhetoric and morality. It is through this third moment that the ontological cartography of the social shifts, that the social world is institutionally territorialized, including here the creation of common markets, regions where such activities take place (Miller and Power 2013).[27] Successful and failed moments of reference are moments affording valuation and devaluation.

We would hypothesize that changing beliefs about and changing effects of these objects would indicate that institutional logics are being politicized and perhaps changed. When money is worthless, when a border does not hold, when a ballot does not count, when people are imprisoned without hearings, when the experimental evidence is irrelevant: these mark anti-institutional

[27] In their overview of the discipline, Miller and Power describe accounting as a "productive force (…) whose technical procedures and forms of calculation are necessarily entangled with institutional aspirations to realize a wide variety of economic ideals and goals" (Miller and Power 2013, p. 687). They identify that accounting practices territorialize by recursively and repeatedly constituting economic spaces and entities. Accounting not only "informs economic decision-making" but constitutes "the domain of economic activity itself" (i.e. individuals, activities and organizations), a process they refer to as "economization" (Miller and Power 2013, p. 579). Our understanding of territorialization is very similar to Miller and Power (2013)'s account of modern accounting's shaping of social life, not unlike how the processes through which the logic of quantification fashioned our societies (Espeland and Stevens 2008). Note however that Miller and Power do not use these value moments, nor do they theorize accounting as an institutional logic.

moments when institutional logics are being challenged, degraded or evacuated. When new objects – electronic tags, cryptocurrencies or detention camps – whose affordances can either claim consonance with these goods or re-specify what those goods "are," these are moments when institutional logics are changing. Territorialization is a primary mechanism in the orders of worth perspective. Boltanski and Thévenot argue that people objectify themselves by bringing objects into play as "evidence" of their worth, instruments of evaluation. But they only do this through acts of valorization of these objects, "endowing them with value" (Boltanski and Thévenot 2006, p. 131) – in French, "en les mettant en valeur" – attributing value and showing value – "éprouver" (testing) meaning to show value. It is only through "valorized objects" that people can fix the nature of a situation and compare it with other situations. Objects need to be valorized to generalize. Valorization is not given by the nature of the objects; it is given by persons' belief in and recognition of an object's ability to objectify a higher common principle. In the former, attributes of the object would be taken into consideration. They are not. So, one must conclude that it is primarily the latter operation that dominates, and hence that subjects' value commitments are critical to the ability of objects to function as instruments of evaluation, as substantiations of particular worlds and the metaphysical principles upon which they are based.

4th Moment: Evaluation. The fourth moment is the moment of evaluation, the establishment of procedures by which the presence or absence, and centrally, the degree of goodness of an object, practice or being is to be known and measured. Evaluation, agreement about the extent to which a person or object participates in that good, is critical to sustaining the good. Evaluation points to the distributive dimension of any value, not only the conditions of access to a value, but the measurement of that value, its more or lessness. Evaluation centers more on quantification than on classification. Here objects function primarily as evidence of worth, or devices by which that worth is measured. It is through evaluation that one sees clearly the internal relation between these moments, specifically with institution and production. Determinations of worth undergird value; conventional representations sustain the realness of this reality. The determination of goodness is part and parcel of the constitution of isness. The temporal continuity of evaluation helps undergird the belief that the value whose variability is represented can be understood to have remained the same, that is one is engaged in a practice of representation of the same thing, that the value is valuable because it is real, that is, because it can be measured and that measurement is socially consequential, not the least of which is in producing hierarchies of access to that good, which secures the power of the powerful whose power derives from that good. Evaluation is a moment where a value's distribution is revealed. It is also a moment in which value can collapse, production halted by a represented valuelessness. We know a value by its absence, by valuelessnes, more fundamental than another value which must be sacrificed, an opportunity cost, which has primacy in Boltanski and Thévenot's understanding of the justification of unequal worth (Boltanski and Thévenot 2006, p. 142). Worth depends on the prospect of worthlessness, and thus always points beyond itself to the moment when questions of evaluation and worth slide towards valuation, towards ontology, towards the uncertainty of both the fact and the value of value. It is at that point that the institutional quality of market mechanisms, for example, become evident.

Evaluation is central to the orders of worth approach. Unlike those who have fled from value for its normative consensualism into impersonal powers, individual interests or cognitive orders, Boltanski and Thévenot make collective participation in and argument over a generalizable value – a common good – manifested through its qualification of persons as worthy, the principle of order of a series of "polities." With its emphasis on justification of states of worth and the qualification of "beings," this moment of evaluation is at the center of their classic text, *On Justification* (Boltanski and Thévenot 2006). Objects here are deployed as an apparatus for the production of evidence, the facts of the case. Theirs is an Aristotelian understanding of justice whose "adequate measure" is "just proportion" between "the quantity of things distributed and the various qualities of persons" (Boltanski 2012, p. 112–113). "Worth is the way in which one expresses, embodies, understands, or represents other people…" (Boltanski and Thévenot 2006, p. 132). Each polity is a state of worthiness organized around a common good which provides the basis of equivalence between beings (Boltanski and Thévenot 2006, p. 77). States of worth are based on a good which constitutes our commonality and thus our capacity to coordinate our actions. For them, valuation is primarily evaluation, the qualification of beings, the ways people have for "assessing themselves and each other" through "tests," particularly of the worthiest, exemplars whose demonstrated generality become "guarantors of the higher common principle" (Boltanski and Thévenot 2006, p. 65, 141).

In the following sections, we will assess the adequacy of our model by re-reading the history of the options pricing model accounted by MacKenzie and Millo (2003). We chose this case for two reasons. First, it provides us with an adequate empirical setting for studying how an institutional object produces the good it embodies – i.e. market value. Second, it is a well-known case study that has been analyzed extensively through a sociology of valuation lens. By adopting an institutional perspective, we seek to unravel the institutional conditions that enable the performativity of market devices – hence addressing a limitation in the sociology of valuation.

Case Vignette: The Institutionalization of Market Value by an Options Pricing Model

1st Value Moment: Institution

The Black-Scholes-Merton model originated not as economic theory; indeed, the original paper was rejected by the academic journals for its lack of economics (Johnson 2015, p. 52). The model was a mathematical engineering of riskless options pricing based on the construction of a continuously adjusted portfolio of underlying assets and cash whose payoff is the same as the option, allowing the price of the option to be derived as the cost of that portfolio adjusted for the variation of the asset price. It was originally practical mathematics built on the assumption of a frictionless market with no transaction costs that did not allow opportunities for arbitrage (Maurer 2002). It instituted a new ontology of risk, one based on implied volatility, versus an actuarial model insurance companies used to value risk. The instituting of an options market was effected through this object, or device, which both depended on and served as an objectification of the new truth of efficient market value. In an efficient market, price variation reflected all available information. Markets had been seen as manipulated; the new theory made the ontology of price seem more efficient and thus more moral.[28] The practices based on the new market mechanism thereby contributed to the persuasiveness of the theoretical postulate of an efficient market. That postulate implied that equity price variation responds to all available information, eliminating arbitrage opportunities.[29] And further it assumed a financialization of the firm, that a firm's equity

[28] Conversation with Daniel Gabay, February 1, 2018, Paris.
[29] The practice – the continuous adjustment of the underlying portfolio – also implied the ontology of "fair value," that market values are, and should be, based on exit prices, a form of accounting practice which was gaining favor during the same period in which Black-Scholes-Merton model gained prominence (Persson et al. 2017).

price was the best indicator of its market value, an option on the value of its assets, and that a firm was and should be most oriented not to maximization of profit, but to the maximization of shareholder value. If these postulates were correct, the new option pricing mechanism based on the model would both reflect and contribute to the efficiency of market pricing, principally by identification of arbitrage opportunities which would be immediately exploited and thereby closed off once identified. The formula assumed a stock price modeled as a log-normal random walk in continuous time, inferred an option price from the cost of the replicating portfolio, and consequently built an equation that allowed one to perfectly hedge any option (MacKenzie 2006, p. 20–21, MacKenzie and Millo 2003, p. 120).

Once the options market was flourishing, options traders using the model did not care about market value: They cared about bid-ask spreads, about deviations from what the model predicts market price should be.[30] That's where money was to be made. Assumptions about the nature of market value were built into the material practices. You did not need to talk about them.[31] Belief in market value was nonetheless critical both in the institution of this practice of options pricing and in sustaining it in the moment of its failure. It was belief in the efficiency of market valuation, that prices equaled market value, that both undergirded the new technical object by which options were priced and overcame opposition by exchange members to its adoption. For those who reconstituted the commodities market to accommodate the inclusion of equity options estimated and actuated in this way, there was both a belief in the transcendental good of market value and in this practice to reveal it. Leo Melamed, a central player in financial markets, whose father had been a socialist Bundist, explained his selfless dedication to this project: "My father had instilled in me [the] idea that you gain immortality by tying yourself up with an idea, or a movement, or an institution that transcends mortality" (MacKenzie and Millo 2003, p. 116). For Melamed, the market offered a transcendental value upon which to elaborate new pricing practices. The initial persuasiveness of options prices derived from the Black-Scholes-Merton model depended on that belief – enabling the gradual shaping of pricing practices through the model.

[30] Ironically, in financial markets it appears that it is the failure of (e)valuation that is necessary to the creation of market value.

[31] Boltanski discriminates between fairness and justice, the first where objects speak for themselves in a "silence of equivalence", "letting themselves being guided by the equivalences that are tacitly enclosed in the silence of existing things" (Boltanski 2012, p. 67).

In 1987's crash, risk became uncertainty, exposing a gap between value and price, or as Eric Rosenfeld worried that day in 1987 about "the end of the whole financial system" (MacKenzie 2006, p. 205). MacKenzie and Millo understand the persistent volatility skew that ensued as lasting traces of a "collective trauma," an experience of a "threat to the very existence of the derivatives market" (MacKenzie and Millo 2003, p. 138). The events that occurred also revealed the way in which belief in market value was critical to the institutional outworking of market pricing in general, and in the sustenance of options pricing as a valid enactment of market value in particular. When market prices plunged, the derivative market was almost destroyed. According to Daniel Gabay, a mathematical economist involved in this area, it was portfolio insurance, the hedging of asset managers replicating the portfolio, that caused the options market to crash, and with it the whole market. Replicating portfolios are built on the assumption you can buy any amount of stock all the time and that the prices move continuously with no jumps. With large price variations you cannot adjust portfolios.[32]

Despite the double failure – of the market for options and its pricing practices, belief in market value, indeed in efficient market value, was not negated, nor was options trading as a way to reveal that value. It was not re-stigmatized as gambling. Rather, in what MacKenzie and Millo term "practitioner know-how" (MacKenzie and Millo 2003, p. 131), actors used different instrumentation – the Cox-Ross-Rubinstein model and Levy distributions, varying practice in order to effect it by market makers setting volatility skews, increasing margin requirements based on a change in functional form from log-normal to a Paretian distribution with "infinite variance" in which the "probability of extreme events is much higher" (MacKenzie and Millo 2003, p. 135), and coordinating moral and coercive pressures on those who sold uncovered puts. "Those who sold such puts without hedging them faced moral obloquy – 'We call them the shit-sellers!'" one interviewee explained (MacKenzie and Millo 2003, p. 135). MacKenzie and Millo argue that the skew is the "upshot of a collective trauma" (MacKenzie and Millo 2003, p. 138). The "moral obloquy" that attached to "shit sellers," those traders who sold uncovered, low strike-price puts, derived from the ways in which their rational actions threatened "the market as a whole" (MacKenzie 2006, p. 208, MacKenzie and Millo 2003, p. 140). The critique returns to the tropes of gambling. The trauma, a contradiction between practice and good, pricing and value, is a kind

[32] Gabay contends that the same thing happened in 2008 with automatic trading in which the mathematical models underlying them still assume you can buy and sell any amount of stock.

of Knightian uncertainty.[33] When price does not index value, the practical prospect of an ontological collapse of market value looms.[34] Belief in market value beyond market pricing was essential here to transform market pricing practices, to remake the market in the image of market value.[35]

2nd Value Moment: Production

The model made options trading legitimate; its generalized use made it a low-cost, high volume, multiply-sited market with minimal transaction costs. It became the vehicle through which the efficient conditions it presumed were produced (MacKenzie 2006, p. 257). As the authors famously show the market converged with the model because members were persuaded to use it in multiple markets, particularly in the identification of spreads in arbitrage trading,[36] and because the conditions assumed by the model – the information technologies, including price sheets and autoquote software, and regulations allowed continuous portfolio adjustment with minimum transaction costs to obtain on the trading floor.[37] The model's fit with actual options prices was only approximate. When the CBOE began trading, options contract prices for calls, for instance, were 30% to 40% overvalued relative to the model, an even worse fit than obtained in the ad-hoc options markets that preceded it (MacKenzie and Millo 2003, p. 121–122). As MacKenzie points out, the identification of risk-adjusted excess returns, non-excessive returns being the core datum of the capital asset pricing model, were undecidable as to whether they should be read as evidence against the model or as evidence against market efficiency (MacKenzie 2006, p. 23). For more than a decade it produced the valuation it modeled.

[33] In economics, Knightian uncertainty is an incalculable risk because it is unknown.
[34] See Maurer (2002) for an in-depth analysis of the role of index in derivatives.
[35] The potential tension between value and price is prominent in Beunza's ethnography of a major investment bank that pioneered highly lucrative interest rate-swap derivatives (Beunza 2019). Although cast in terms of the amoralizing effects of the models, Beunza's work shows the importance of alternative ontologies of market value, and their potential independence of price, the ways in which these ontologies are afforded by new market devices, and the importance of belief in sustaining the device-ontology couplets. The politics and apparently amoral perversities of valuation are grounded here, not in a group culture.
[36] The offsetting selling of overpriced options and buying underpriced one.
[37] MacKenzie and Millo also point to the role of trader identification with the analytic practices (MacKenzie and Millo 2003, p. 125).

3rd Value Moment: Territorialization

The Black-Scholes-Merton options pricing model was part of a change in the ontology of market value. Options trading is driven primarily not by valuations of the firm whose price variations they hedge, but by the valuation of the riskiness of those firm valuations, by the volatility that the option price implies (MacKenzie and Millo 2003, p. 125–126). "Gradually, what was being bought and sold in an options market was reconceptualized: it was the Black-Scholes-Merton model's key parameter, *volatility*." (MacKenzie and Millo 2003, p. 125). The authors point to the ways in which the pricing model "necessary for risk-management… offered a way of communicating and coordinating activities, of *talking* about options." The performativity of the Black-Scholes-Merton model, where the objectification and deployment of representation created the real, was institutionally constrained and conditioned.[38] There is a historicity to it. The very emergence of this form of options trading illustrates the historically contingent relation between market value and price, here the institutional fact that prices were, for a considerable period, not market values at all. On the one side, the Supreme Court had ruled in 1905 that in the absence of an underlying physical asset through which trading could be settled, futures trading was gambling, not a legitimate form of trading (MacKenzie and Millo 2003, p. 113, Millo 2007).[39] On the other, the collapse of Bretton-Woods with Nixon's decoupling of the dollar and gold in 1971 eroded the ability of money to function as a metric of market value given the variable exchange rates and increased volatility of asset prices in its wake (Johnson 2015, p. 52). It also results from an historical account of mathematics and the role of probabilities in the latter (Maurer 2002). Maurer (2002) indeed argues that the discursive power of derivatives (i.e. option pricing theory) notably finds its origins in the separation between religion and the mathematical models used in those financial techniques.[40] The emergence of this new institutional object depended on a new moral evaluation of the market, the performative power of the state, and the erosion of another institutional object. It also depended on an entire material infrastructure – the computer and the

[38] It is not, strictly speaking, an Austinian performative speech act, in which the saying is the doing.
[39] Before the Chicago market instituted market trading in stock options, they were traded over the counter.
[40] "Derivatives' indexical power," Maurer explains, "is a symptom of this trauma." (Maurer 2002, p. 15). According to Maurer, the separation between subjects, objects and relations between both poses a fundamental problem as it assumes a correspondence between a fact and a sign, anteriority and referentiality. He concludes: "The mathematical black boxes of financial derivatives are a moral argument." (Maurer 2002, p. 31).

internet – that provided the instantaneous information and the credit to continuously adjust the underlying portfolios.

4th Value Moment: Evaluation

In the pricing of options, it is less the expectations of profit than the evaluation of price risk that is central to valuation. Risk is integral to valuation. Failure is the outside inside of risk, the limit and the ground of value. Risk points to the zero term which must ground every evaluation. Like death and bankruptcy, risk points to the potential failure of value, to its absence, its erasure.[41] Option pricing registers expectations about changes in value; it can also express the prospect that price does not reflect value, that prices are wrong, in the worst case that they may not be simply unreliable or biased, but invalid. It is when prices are worthless that belief in market value is most critical. When the stock market crashed in 1987, the efficient market almost destroyed the options market and its clearing system. Put options prices and their implied volatilities jumped 800%. The Black-Scholes-Merton valuation device failed. The flat line resulting from using the log normal model between strike price and implied volatility disappeared, never to return (MacKenzie 2006, p. 200–203, MacKenzie and Millo 2003, p. 128–130). Observed options prices no longer conformed to the model, particularly puts with low strike prices relative to the indices. Implied volatility has subsequently remained skewed, a skew inconsistent with the premise of an efficient market assumed by the device, even greater than can be accounted for by the actual distribution of prices (MacKenzie 2006, p. 204). The market now anticipates disaster, discontinuous price

[41] Risk, and risk management in general, is not only a form of valuation of outcomes, it depends on them (Douglas, 1990). Uncertainty is the limit of risk. For Knight (1921), risk is a situation where uncertainty is measurable. Uncertainty, both known and unknown, is primordially institutional. It is this horizon of uncertainty that makes belief essential to the sustenance of value. It is when risk verges on worthlessness that we understand the theoretical limitations of Dewey's turning the question of value into a verb form alone, as valuation (Hutter et al. 2015, Muniesa 2011). Hutter, Stark and Berthoin Antal (2015), following Dewey, argue that we should displace value by valuation, the noun by the verb. Dewey wrote: 'there are no such things as values. . . There are things, all sorts of things, having the unique, the experienced, but undefinable, quality of value' (Dewey 1923, p. 617) cited in Muniesa (2011, p. 25). It is risk that provides an opening for understanding the ways in which actors consider value as a noun, a non-phenomenal belief in a "thisness" that is the basis of the institutional logic of market value. And it also opens the prospect that this experience of the institutional is grounded in the real, that the logic of the institutional world must be like this for such experiences to animate and sustain forms of practice. Values are not entities or things, but the reproduction of action depends on actors believing that they subsist independently of that action, and not just as subjective states which they attach to the material world as objects of knowledge and desire.

movements and dramatic shifts in volatility (MacKenzie 2006, p. 207). The American options markets resemble markets for catastrophic risk, like insurance, not equities. With its single-minded focus on the volatility skew, option markets now price failure. The skew represents an anticipation of failure, of valuelessness, of the probable improbable limit. Failure is the bright shadow of value.

The Co-Constitution of the Four Value Moments of the Options Pricing Model

The emergence of options trading on the Chicago exchange was not just a case of the politics of reference, to which objects market value can refer, but part of a broader ontological political shift, a transformation of what market value is and how it is produced, here through the electronic deployment of a new mathematical equation deployed as a modality of efficient pricing. The deployment of the Black-Scholes-Merton options pricing device involved the four value moments described in our theoretical model. The institutionalization of options pricing exemplifies their co-implication, their constellation as a relational unity. They operate more like a production function or a practical structure, than a set of independent variables whose effects can be netted out and isolated as conditional probabilities. The introduction of this new device per se did not dictate that it would successfully become part of a new material routine for market pricing. Its institution, its use as an iterable model, depended on a belief that it was consistent with, indeed productive of, market value. The device materialized a particular ontology of market value of the firm – both its ground and its temporality, financialization of the firm, understood as a legally bounded aggregation of fungible assets, in which shareholder value, not the profitability of material assets, on the one side, and accounting based on exit prices, on the other side, were the bases of a firm's market value. The model offered a financialized form of producing and evaluating value both by reducing option prices thereby increasing the expected profitability of owning – and selling – them and by offering a more reliable identification of arbitrage opportunities as a difference in returns between different financial instruments (MacKenzie 2006, p. 257). And it transformed the domain of reference of the market by allowing options to be traded, legitimating the value of market misvaluation as a priced market object (cf. Figure 3).

Insert Figure 3 about Here

Conclusions

In this essay, we have explored the role of institutional objects in the institution, production, evaluation and territorialization of the good. We decompose those practices into four value moments through which the good that undergirds an institutional logic is objectified. Largely bracketing subject and subjectification, we focus on practices of objectification of the good. Over the past years, there have been several calls to address the role of objects in institutional logics (Jones et al. 2013, 2020). Our model points to the constitutive role – both constituting and constituted – of a particular kind of object – institutional objects, objects which depend upon particular ontologies of the good. It is these goods – as objects and objectives – which anchor an institutional logic. Institutional objects fuse fact and value through four co-constitutive value moments: the institution of the good, its production, its evaluation and its territorialization. If our conceptualization of institutional logics as grammars of valuation has some validity, we should be able to study the specific ways in which institutional objects are deployed and appropriated across moments of valuation – institution, production, evaluation and territorialization – the consonances and tensions between these deployments and their role in institutional persistence and change. This has a number of implications.

First, it suggests that institutional logics can be studied as grammars of valuation practices. We believe this approach is consonant with the way in which institutional logics must be observed in practical movement and yet be grounded in non-phenomenal goods which are treated by participants as if they were real and invariant. We believe that the coherence of this grammar is part of what sustains an institutional logic. But such a grammar also identifies the possibilities and locations for change, break-down, and political challenge and innovation. There is nothing static about an institutional logical approach. It also suggests a bridge between practice-centered theories which have centered on ordinary, day-to-day practices and the institutional logical approach which has centered on epochal, societal transformations or shifting institutional cartographies. These value moments are found everywhere.

Second, methodologically, our framework implies that institutional logics can be studied through the duality of objects and practices of valuation. The general possibility of linking the material and the symbolic was latent, but unexplored, in the original conceptualization of institutional logics as "sets of material practices and symbolic constructions" (Friedland and

Alford 1991, p. 158). It is through the conjunction of material objects and metaphysical goods that one can identify the institutional quality of practice, neither reducible to materiality nor meaning, fusing instrumental and value rationality. And it is through this doorway that we might restore value to social practice, not as a subjective commitment and belief, but as constitutive of the *agencement* (Callon et al. 2007, Callon and Muniesa 2005) of an operative assemblage of practice.

Third, institutional logics depend on constellations of objects, both institutional and non. We assume that an institutional logic is attached to a constellation of institutional objects, not unlike the way Foucault understood the productive constellations of governmentality and bio-power (Foucault 1980, 1991). The production of goods, such as market value depends on practices of priced exchange conducted through constellations of interlocked – inter-objective Latour would say – institutional objects – money, contract, accounts, corporation, account, equity, as well as objects which have no internal relation to the goods that ground its practices, but are claimed and organized by them nonetheless. We have not conceptualized the relationships between institutional objects themselves within a logic or their articulation where a plurality of logics is at play. So, we are not yet in a position to consider cases such as Fourcade (Fourcade 2011a)'s study of the valuation of nature, and how market devices articulate with 'scientific' institutional objects, such as those used by conservationists to estimate biodiversity.

Fourth, the identification of the four co-constitutive value moments as the mechanisms through which institutional objects produce goods implies that if one moment is missing, then, objects are unlikely to become institutional. For instance, in the case of the adoption of ESG criteria by financial actors (Beunza and Ferraro 2019), the institutional meaning of market value is not likely to integrate social and environmental concerns if this is not accompanied by a belief in the adequacy of the ESG criteria, their use to produce market value, the sharing of common reference and their shared operationalization as evaluation criteria. We believe that our theoretical model could help understand why some market devices help transform practices towards the purposes they pursue, while others do not. At a time when rankings, reporting and performance systems abound, often with controversial outcomes, uncovering the institutional foundations of such market devices is certainly worth exploring, both for institutional scholars and sociologists of valuation (see Arjaliès Forthcoming for an application of the theoretical framework developed in this essay).

And finally, there is a useful, consonance between a religious sociology and this institutional logical view. An institutional object hinges on the subjectification that complements the objectification that we have made central in this essay. The goods of an institutional object can never be seen; through their visible icons they call us to projects and their practices which author and authorize us, a material objectification that affords the production of invisible goods at which we can never arrive. Institutional objects are productive symbols, anonymously authored icons of the goods, icon in the sense of an image of an invisible God, an invisibility and unknowability that is the icon's condition of possibility, a visibility that depends upon and gives access to that invisible unknown, like the body and image of Jesus does to God (Conty 2009). In the Greek tradition, those objects which have been cut off from their invisible reality are "unsubstantial," in a word, *eidolon*, idols (Conty 2009, p. 6). In parallel with the way in which Marion (1991, p. 10–11) distinguished between idol and icon, the idol made by our gaze, an object that objectifies our intention, and an icon, that which provokes our gaze, generating an intention whose object must necessarily be invisible, institutional logics subjectify us through our being seen and intended through these material practices and the goods that ground them. Institutional objects are icons whose institutional meaning depends upon their deployment in valuation process, not idols located in individual selves. Their objectifications typically impose constraints on individuals' behaviors such that gaps between idol and icon do not emerge in public, and thus the possibility of contest and critique is occluded. Institutional life simultaneously subjectifies and objectifies. Given the material immanence it requires, it is never iconoclastic. Given its dependence on non-phenomenal goods, if it is idolatrous, reducing its account or its engagement to its visible material practices alone, it risks the death of its goods.

Conceptualizing, measuring and specifying the relationship between different kinds of objects, the practices they afford, and the goods that ground them suggests that we explore the ways in which this approach may or may not articulate with the assemblages of actor-network theory, the intra-active apparatus of agential realism, the sites of practice in practice theory, and the formats and equipment in orders of worth. We understand that we are very far from our objective in part because we do not yet understand the productivity of our conceptual objects, to what kinds of things they can refer, and how they will be evaluated in giving us the whatness of what is.

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LIST OF FIGURES

Figure 1: An Institutional Logic

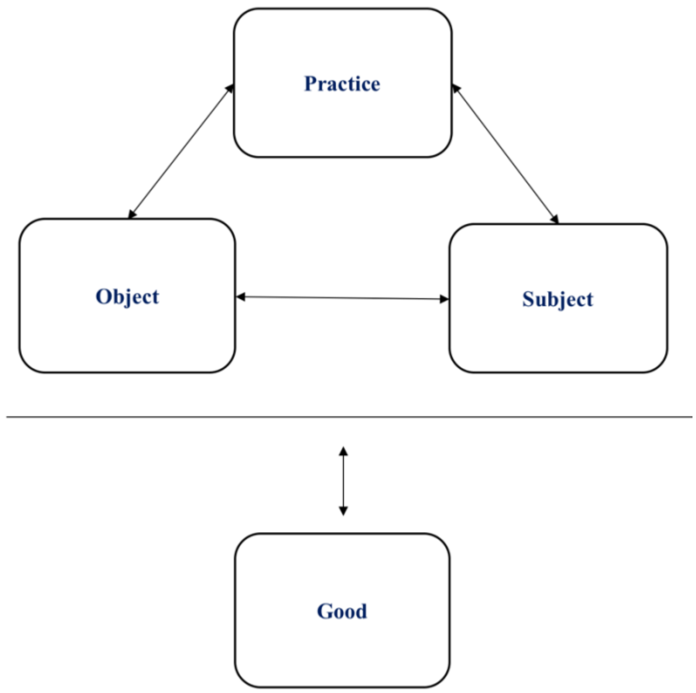

Figure 2 – The Institutionalization of Good through the Four Value Moments

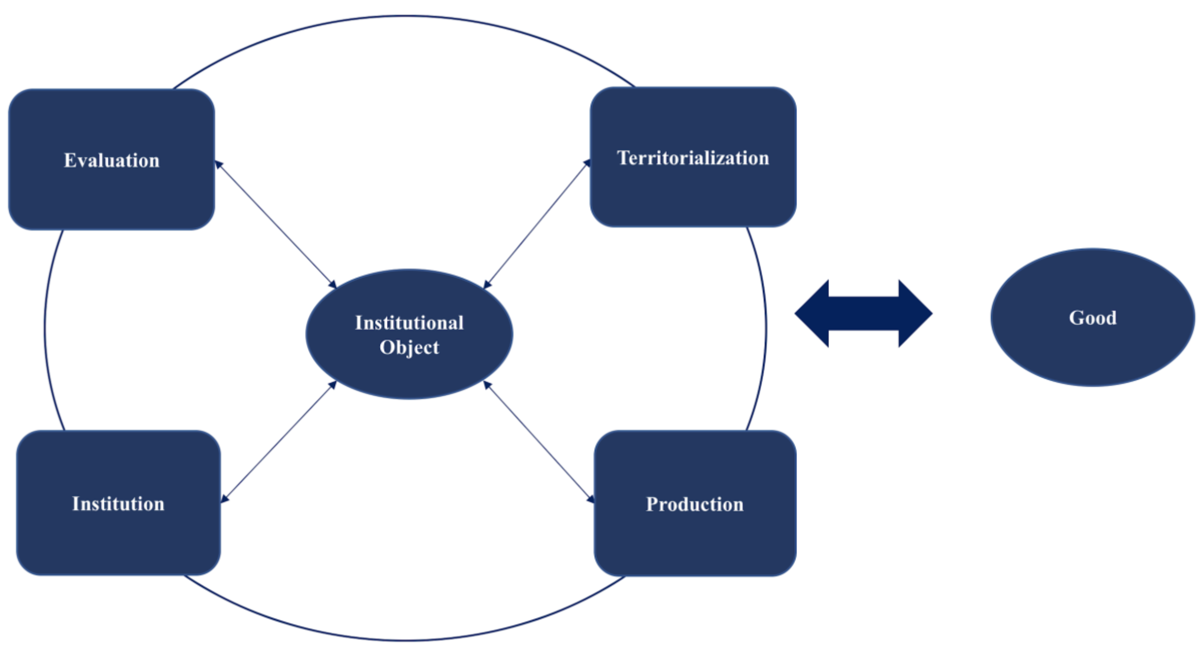

Figure 3 – The Institutionalization of Market Value through the Options Pricing Model